

MedOrchestra — Product Requirements Document

Project Type: B.Tech Final Year Capstone (CSE — Health Informatics)

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Reference Paper: Borkowski et al., "Multiagent AI Systems in Health Care", PMC12360800 (2025)

1. Executive Summary

MedOrchestra is a multi-agent AI clinical decision support system (CDSS) that implements and extends the theoretical framework proposed by Borkowski et al. (2025). The system orchestrates four specialized AI agents — Triage, Diagnostician, Drug Safety, and Critique — using LangGraph to simulate a real hospital clinical reasoning workflow. It is the first working prototype combining stateful multi-agent orchestration, Graph-RAG over drug interaction data, and an adversarial critique mechanism in a health informatics context.

The primary academic contribution is the **Critique Agent**: an adversarial agent that challenges diagnostic outputs for low-confidence reasoning — a concept absent from all known prior implementations in this domain.

2. Problem Statement

2.1 Clinical Context

Diagnostic errors account for approximately 40,000–80,000 deaths annually in the US alone. Existing AI-based CDSS tools are monolithic — a single model produces a diagnosis with no internal verification, no source citations, and no adversarial checking of its own reasoning. They also fail to account for drug-drug interactions during the diagnostic process.

2.2 Research Gap

Borkowski et al. (2025) proposed a 7-agent multi-agent framework for sepsis management but left it entirely theoretical with no implementation. All existing GitHub projects in this space are either:

- Simple RAG chatbots without true agent orchestration
- Conceptual frameworks with no data pipelines
- Benchmarking tools without clinical workflow modeling

No existing system combines: LangGraph-based stateful orchestration + Graph-RAG over drug interaction knowledge graphs + an adversarial critique agent.

3. Goals & Success Criteria

3.1 Academic Goals

- Demonstrate a working 4-agent pipeline on synthetic patient cases
- Achieve $\geq 75\%$ diagnostic accuracy on MIMIC-III-derived test cases
- Produce a publishable paper framed as: *"First working prototype of the Borkowski et al. framework with novel adversarial critique mechanism"*
- Submit to a relevant venue (e.g., AMIA, IEEE CBMS, or arXiv pre-print)

3.2 Technical Goals

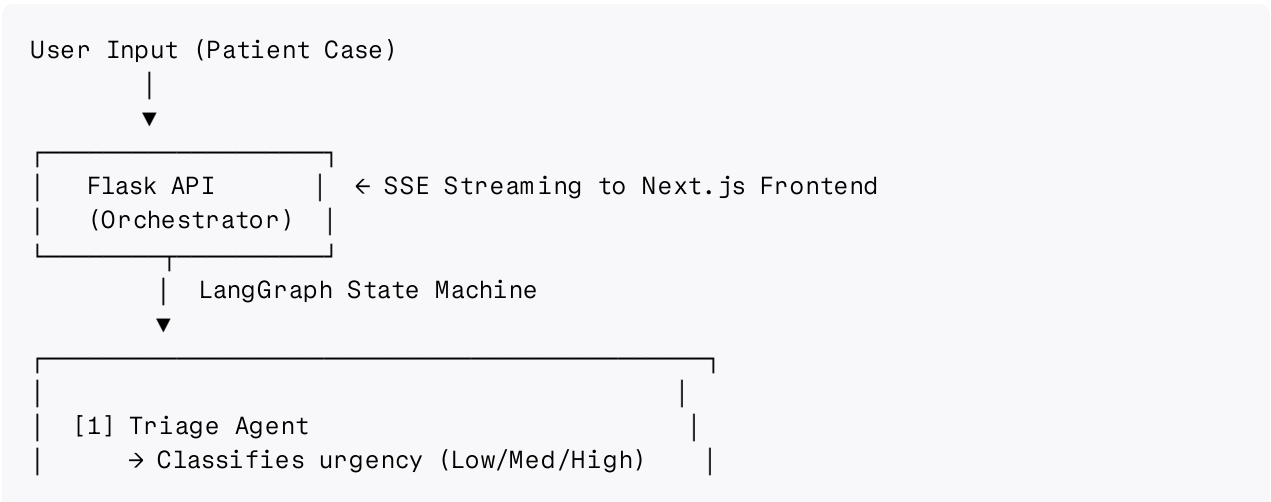
- End-to-end latency under 15 seconds for a complete 4-agent reasoning pass
- RAG retrieval precision ≥ 0.70 on held-out PubMed queries
- Drug interaction detection recall ≥ 0.80 on DrugBank test pairs
- Full streaming UI with real-time agent reasoning visible to user

3.3 Out of Scope

- Real EHR/FHIR integration (future work)
- Deployment to actual clinical environments
- HIPAA compliance (system uses synthetic/open-source data only)
- More than 4 agents (scope is intentionally constrained for timeline)

4. System Architecture

4.1 High-Level Overview



5. Tech Stack

Component	Technology	Justification
Agent Orchestration	LangGraph (LangChain)	Stateful graph execution, native streaming
LLM Inference	Groq API (llama3-70b-8192)	Free tier, ultra-fast inference, no GPU needed
Embeddings	BAAI/bge-small-en (HuggingFace)	Free, high quality, runs on CPU
Vector Store	Supabase pgvector	Free tier, already familiar, SQL-based
Drug Graph	NetworkX + DrugBank CSV	Open-source, no paid API
Backend API	Flask + SSE (Server-Sent Events)	Lightweight, familiar, streaming support
Frontend	Next.js 14 (App Router)	Familiar stack, real-time streaming UI
Data Sources	PubMed OA Subset + ICD-11 + DrugBank	All free and open-access
Evaluation	MIMIC-III synthetic cases	Open-access clinical benchmark
Dev Environment	VS Code + Google Colab (indexing)	Existing setup

6. Data Sources & Pipelines

6.1 PubMed Open Access Subset

- **Source:** https://ftp.ncbi.nlm.nih.gov/pub/pmc/oa_bulk/
- **Scope:** ~500K abstracts filtered to internal medicine, emergency, pharmacology
- **Chunking:** 512 tokens with 64-token overlap
- **Embedding:** BAAI/bge-small-en (384-dim vectors)
- **Store:** Supabase pgvector table pubmed_chunks

6.2 ICD-11 Ontology

- **Source:** WHO ICD-11 API (free)
- **Use:** Structured disease code lookup to ground diagnoses in standard terminology
- **Integration:** Post-process Diagnostician output to attach ICD-11 codes

6.3 DrugBank (Drug Interaction Graph)

- **Source:** DrugBank Open Data CSV (free academic license)
- **Structure:** NetworkX directed graph — nodes = drugs, edges = interactions with severity
- **Graph-RAG:** Given a medication list, run graph traversal (BFS depth-2) to find all interaction pairs
- **Schema:** {drug_a, drug_b, severity: [minor/moderate/major], mechanism, evidence_pmid}

6.4 Evaluation Dataset

- **Source:** MIMIC-III Demo (free, no IRB needed for demo version)
- **Use:** Generate 50 synthetic patient cases for evaluation
- **Metrics:** Diagnostic accuracy, drug interaction recall, end-to-end latency

7. Agent Specifications

7.1 Triage Agent

Attribute	Detail
Input	Patient complaint text + vitals (HR, BP, SpO2, Temp)
Output	Urgency level: LOW / MEDIUM / HIGH + reasoning
Method	Groq zero-shot classification with structured output (JSON mode)
Prompt	System: Clinical triage nurse. Classify urgency based on SEWS scoring guidelines.
Fallback	Rule-based vital sign thresholds if LLM fails

7.2 Diagnostician Agent

Attribute	Detail
Input	Triage result + patient complaint
Output	Top 3 differential diagnoses, each with confidence (0-1), ICD-11 code, and 2-3 PubMed citations
Method	RAG: embed query → retrieve top-5 PubMed chunks → augmented generation via Groq
Retrieval	Cosine similarity search on Supabase pgvector, threshold ≥ 0.72
Prompt	System: Internal medicine specialist. Use retrieved evidence only. Cite sources.

7.3 Drug Safety Agent

Attribute	Detail
Input	Patient medication list + diagnoses from Agent 2
Output	List of drug-drug interaction alerts with severity and mechanism
Method	Graph-RAG: extract drug entities → NetworkX BFS traversal → Groq for plain-English explanation
NER	spaCy en_core_sci_sm for drug entity extraction from medication list
Prompt	System: Clinical pharmacist. Explain each interaction in plain English for a clinician.

7.4 Critique Agent (Novel Contribution)

Attribute	Detail
Input	All outputs from Agents 1-3 + full reasoning chain
Output	Confidence score (0-1), flagged weaknesses, final clinical summary
Method	Groq structured critique prompt — acts as "devil's advocate"
Re-route Condition	If confidence < 0.60, triggers one re-route to Diagnostician with specific clarification request
Prompt	System: Senior attending physician reviewing a junior's differential. Challenge unsupported claims. Flag if diagnoses lack evidence. Produce final confidence score.

8. Frontend Specifications

8.1 UI Layout

MedOrchestra [Logo] [Dark Mode]

Patient Input Panel

Chief Complaint: [text area]

Vitals: HR [] BP [] SpO2 [] Temp []

Current Medications: [tag input]

[Analyze Case →]

Agent Reasoning Timeline (SSE Streaming)

Triage Agent

[HIGH urgency] ✓

Diagnostician

[thinking...] ⌛

Drug Safety

[waiting]

Critique

[waiting]

Final Clinical Report Panel

(rendered after all agents complete)

8.2 Streaming Architecture

- Flask backend uses text/event-stream SSE
- Each agent emits events as it starts, thinks, and completes
- Next.js frontend uses EventSource API to render streaming updates
- Agent timeline animates in real-time — each card expands with output as it arrives

8.3 Final Report Format

The rendered final report includes:

- Urgency Banner (color-coded: green/yellow/red)
- Differential Diagnoses — ranked list with confidence bars and PubMed citations
- Drug Interaction Alerts — severity badges (Minor/Moderate/Major)
- Critique Summary — overall confidence score + any flagged concerns
- Export Button — downloads report as PDF

9. Evaluation Plan

9.1 Quantitative Metrics

Metric	Target	Measurement Method
Diagnostic Accuracy (Top-1)	≥ 65%	MIMIC-III synthetic cases, ground truth ICD codes
Diagnostic Accuracy (Top-3)	≥ 80%	Same as above
Drug Interaction Recall	≥ 80%	DrugBank held-out interaction pairs
RAG Retrieval Precision	≥ 70%	Human-annotated PubMed relevance labels
End-to-End Latency	≤ 15s	Timed on 50 test cases
Critique Confidence Calibration	Pearson $r \geq 0.6$	Critique score vs. human physician rating

9.2 Qualitative Evaluation

- 2-3 case studies with detailed walkthrough of agent reasoning
- Comparison table: MedOrchestra vs. GPT-4 direct diagnosis vs. rule-based CDSS baseline
- Ablation study: system performance with/without Critique Agent

10. Project Timeline

Week	Milestone
Week 1-2	Set up Supabase, embed 50K PubMed abstracts, test retrieval
Week 3-4	Build Triage + Diagnostician agents, LangGraph skeleton
Week 5-6	Build DrugBank graph, Drug Safety Agent with Graph-RAG
Week 7-8	Build Critique Agent + re-routing logic, full pipeline integration
Week 9-10	Flask SSE API, Next.js frontend with streaming timeline UI
Week 11	Evaluation on MIMIC-III cases, metric collection
Week 12	Paper writing (Introduction, Methodology, Results)

Week	Milestone
Week 13	Paper writing (Discussion, Conclusion, References), submission prep

11. Novel Contributions Summary

- 1. First working implementation of the Borkowski et al. (2025) multi-agent health informatics framework
- 2. Critique Agent with feedback loop — adversarial agent that re-routes low-confidence diagnoses (not present in any prior work)
- 3. Graph-RAG over DrugBank integrated into a clinical reasoning pipeline (not done in existing MAS healthcare projects)
- 4. Streaming multi-agent UI — real-time visibility into agent reasoning for clinical transparency
- 5. Open-source release — full codebase and evaluation dataset released on GitHub for reproducibility

12. Risks & Mitigations

Risk	Likelihood	Mitigation
Groq API rate limits during evaluation	Medium	Batch test cases, add retry logic, cache responses
PubMed retrieval quality too low	Medium	Tune chunking strategy, increase retrieval k, add BM25 hybrid search
DrugBank academic license delay	Low	Use FDA drug interaction database as fallback (public domain)
LangGraph re-routing creates infinite loops	Medium	Hard cap: max 1 re-route per case
MIMIC-III access approval delay	Low	Use MIMIC-III Demo (no approval needed) for initial evaluation

13. Paper Outline (Target: IEEE CBMS 2027 or AMIA 2027)

- 1. Introduction — Clinical need for transparent multi-agent CDSS; gap in Borkowski et al.
- 2. Related Work — Single-agent CDSS, LLM-based diagnosis, RAG in healthcare, MAS frameworks
- 3. System Design — Architecture, agent specifications, LangGraph state machine
- 4. Implementation — Data pipelines, Graph-RAG, streaming API
- 5. Evaluation — Metrics, MIMIC-III results, ablation study
- 6. Discussion — Critique Agent impact, limitations, clinical applicability
- 7. Conclusion & Future Work — FHIR integration, more agents, federated learning

